

Project Executable Files

Date	06 July 2026
Team ID	SWTID-2026-8308
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example similar) or	Yes
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA

8	Dockerfile / Containerization config (if applicable)	NA
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9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

/OptiCrop

— /frontend	- User interface (HTML, CSS, JavaScript/React)
— /backend	- Server-side application and APIs
— /models	- Machine Learning models for crop prediction
— /database	- Database schema and SQL scripts
— /datasets	- Crop, soil, weather and fertilizer datasets
— /config	- Configuration and environment files
— /uploads	- User uploaded images and files
— /reports	- Generated analysis reports
— /tests	- Unit and integration test files
— /docs	- Project documentation
— README.md	- Project overview and setup instructions
— requirements.txt	- Python dependencies
— package.json	- Frontend dependencies (if applicable)
— .env.example	- Environment configuration sample
— app.py / main.py	- Main application entry point
— LICENSE	- License file (optional)

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://ashadeepaanusuri.pythonanywhere.com
Login Credentials (Demo)	[Provide demo username and password for evaluator access]
Platform / Hosting Provider	pythonanywhere
Repository Link	https://github.com/ashadeepaanusuri-codes/OptiCrop--Smart-Agricultural-Production-Optimization-Engine-

Demo Video Link	https://drive.google.com/file/d/1F9WYG1dns7xFmOMZ50XUHKvrgelrSrB/view?usp=drivesdk
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Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

Operating System: Windows 10/11 or Linux

Python 3.10 or above

MySQL 8.0 (or SQLite if applicable)

Visual Studio Code (Recommended)

Git (Optional)

Required Python packages listed in requirements.txt

Step 1:

Download or clone the OptiCrop project source code to your local system.

Step 2:

Install all required dependencies using:

S.No	Known Issue / Limitation	Workaround / Status
1	Crop recommendations depend on the quality and accuracy of the input soil and weather data.	Users should provide accurate and updated data. Future versions will integrate real-time IoT sensors and weather APIs.

2	The system requires an active internet connection to access external weather	Basic features work offline, while online features require internet
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pip install -r requirements.txt

Step 3:

Configure the environment by copying .env.example to .env and updating the database credentials if required.

Step 4:

Create the database and import the provided SQL schema (opticrop.sql).

Step 5:

Start the application using:

python app.py

(or python main.py, depending on your project)

Step 6:

Open a web browser and access the application at:

http://localhost:5000

(Use http://localhost:8000 if your project runs on port 8000.)

Step 7:

Log in using the demo credentials and begin using the OptiCrop: Smart Agricultural Production Optimization Engine to analyze soil data, receive crop recommendations, optimize production, and generate reports.

Step 5: Known Issues / Limitations

	information and online services.	connectivity.
3	The current model supports a limited set of crops and regional datasets.	Future updates will include additional crops, regions, and improved machine learning models for higher prediction accuracy.